

	$v = 4.51 \times 10^{-4} \text{ m s}^{-1}$
6(a)	Due to wave-particle duality, the electrons possess wave properties with de Broglie wavelength $\lambda = \frac{h}{p}$. Therefore they experience diffraction when passing through the graphite film and interference after passing through the graphite film, resulting in concentric rings on the fluorescent screen. The bright rings are due to constructive interference and the dark rings are due to destructive interference.
6(b)	With a higher p.d. applied, more electric potential energy is converted to kinetic energy of the electrons. The electrons therefore move faster and possess larger momentum. This results in them having smaller de Broglie wavelength. The spacing between the bright and dark interference rings therefore decreases. More interference rings can also be observed.
6(c)	Gain in kinetic energy = loss in electric potential energy $\frac{1}{2}mv^2 = eV$ $\frac{1}{2}(9.11 \times 10^{-31})v^2 = (1.6 \times 10^{-19})(1200)$ $v = 2.053 \times 10^7 \text{ m s}^{-1}$ de Broglie wavelength $\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{(9.11 \times 10^{-31})(2.053 \times 10^7)} = 3.545 \times 10^{-11} \text{ m}$

7(a)(i)	With a glide ratio of 40, a 1 m vertical drop results in a 40 m horizontal distance
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